MOHAMMAD ABDUL MOHAIMIN

Mechanical Engineer — Plant Inspection, Reliability & Asset Integrity

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PROFESSIONAL SUMMARY

Mechanical Engineering graduate from Bangladesh University of Engineering and Technology (BUET) with academic and professional experience spanning **mechanics of materials, composite materials, structural analysis, and mechanical integrity**. My undergraduate thesis focused on the **multiphysics behavior of hybrid laminated composites**, providing experience in mathematical modeling, numerical analysis, and computational simulation. Through academic projects and my current work in refinery inspection and mechanical integrity, I have gained practical exposure to **material degradation, corrosion, failure mechanisms, structural assessment, and the behavior of engineering materials under different operating conditions**. These experiences have strengthened my interest in understanding the mechanical behavior and performance of materials and structures. I am eager to pursue graduate study to deepen my theoretical and computational knowledge, develop stronger research skills, and explore challenging problems in mechanics and materials.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Mechanical Engineering **Feb 2020 – Mar 2025**

CGPA: 3.30 / 4.00

Relevant Coursework: Mechanics of Solids, Numerical Analysis, Metallic Materials, Mechanics of Machinery, Machine Design, Composite Materials, Automobile Engineering.

Research Interests: Mechanics of Materials, Computational Mechanics, Composite Structures, Numerical Modeling, Nanomaterials, Material Failure Analysis.

Chittagong Collegiate School & College

Higher Secondary Certificate (HSC) **2019**

Chittagong Model School & College

Secondary School Certificate (SSC) **2017**

UNDERGRADUATE THESIS

Multiphysics Modeling of Hybrid Laminated Composite Conductors Under a DC Field

Undergraduate Thesis **Nov 2023 – Mar 2025**

Supervisor: Dr. Shaikh Reaz Ahmed, Professor, Dept. of Mechanical Engineering, BUET

Developed a semi-analytical numerical framework to solve the fully coupled electro-thermo-mechanical problem in particle-reinforced hybrid laminated composites.

- Formulated the governing equations and a numerical model for hybrid laminated composite conductors, introducing a Displacement Potential Function approach that reduces the three-variable mechanical equilibrium equations to a single, solvable fourth-order partial differential equation.
- Implemented the Finite Difference Method (FDM) with iterative solvers to solve the fully coupled electro-thermo-mechanical system, incorporating non-linear, temperature-dependent material properties for realistic stress prediction.
- Developed and optimized computational code in MATLAB, validating voltage, temperature, and stress-profile results against industry-standard FEA software (COMSOL Multiphysics) to establish model credibility.
- Core expertise applied: numerical methods (FDM, iterative solvers), Multiphysics modeling, classical lamination theory, heat transfer, and MATLAB programming.

Supervisor: Dr. Shaikh Reaz Ahmed, Professor, Department of Mechanical Engineering

Abstract: Modern electronics, aerospace systems, and high-power applications increasingly require conductors that combine superior electrical conductivity, high thermal dissipation capability, and structural mechanical integrity. In this undergraduate thesis, a semi-analytical multiphysics modeling framework was developed to predict the coupled electro-thermo-mechanical response of hybrid particle-reinforced laminated composite conductors under steady DC current. A Displacement Function (ψ) formulation was introduced to reduce the coupled 2D mechanical equilibrium equations to a single fourth-order partial differential equation, eliminating the need to solve coupled displacement components simultaneously. Temperature-dependent electrical resistivity and thermal conductivity were incorporated with non-linear Joule heating. The governing equations were discretized using a finite difference network (401 × 101 mesh) and solved iteratively. The model was applied to Al-SiC and Al-Graphene symmetric laminates with continuous power-law volume fraction gradation ($V_f = m \cdot x^n$) and intermediate bonding layers (Silver Filled Epoxy, Graphene-Epoxy, Silver Nanowire Paste). Predictions were validated against Finite Element Method (COMSOL Multiphysics) simulations, demonstrating excellent agreement with over 85% reduction in computational effort compared to fine FEA meshes.

ACADEMIC ENGINEERING PROJECTS

Design, Fabrication, and CFD Analysis of a Double-Pipe Heat Exchanger with Disc-Type Turbulator and Outside Fins

ME 310: Heat Equipment Design Sessional, Department of Mechanical Engineering, BUET | Jul 2023 – Sept 2024

Designed, fabricated, and computationally analyzed a 1-metre counter-flow double-pipe heat exchanger incorporating a 13-disc perforated turbulator and 8 external longitudinal fins for passive heat-transfer enhancement.

- Performed MATLAB-based parametric and iterative NTU-effectiveness analysis to optimize heat-exchanger geometry and fin density for improved thermal and hydrodynamic performance.
- Conducted 3D CFD analysis in SimScale using a 727.7k-cell hexahedral mesh to evaluate flow mixing and heat-transfer enhancement produced by the turbulator.
- Demonstrated an additional 2 °C temperature drop attributable to the turbulator through comparative CFD analysis.
- Fabricated the mild-steel prototype in the departmental machine shop and successfully verified its structural and pressure-boundary integrity through a 400 psi hydrostatic pressure test.

Autonomous Detection of Potential Mosquito Breeding Sites Using UAVs and Computer Vision

ME 366: Instrumentation and Measurement Lab, Department of Mechanical Engineering, BUET | Nov 2022 – Mar 2023

Designed, fabricated, and field-tested a custom quadcopter UAV integrated with an onboard camera, GPS telemetry, and a deep-learning computer-vision system for autonomous detection of potential mosquito breeding sites.

- Designed and assembled a functional UAV platform using flight controllers, BLDC motors, ESCs, telemetry systems, and transmitter-receiver modules.
- Developed an automated aerial image-acquisition pipeline from UAV video feeds for real-time environmental monitoring.
- Trained and implemented a YOLOv5 object-detection model on a custom dataset to identify stagnant water and potential breeding sites in difficult-to-access urban locations.
- Integrated GPS telemetry to geolocate and log identified breeding sites, enabling targeted intervention and more efficient public-health surveillance.

Mitigation and Utilization of Methane Emissions from Landfills of Dhaka: Waste-to-Energy Feasibility Study

ME 413: Energy and Environment , Department of Mechanical Engineering, BUET | 2023 – Dec 2024

Conducted a comprehensive environmental and thermo-chemical feasibility study of methane recovery and waste-to-energy conversion from Dhaka's major landfill sites, Matuail and Aminbazar.

- Applied the U.S. EPA Landfill Gas Emissions Model (LandGEM v3.02) to estimate long-term methane generation using Bangladesh-specific waste and landfill parameters.
- Modeled methane-generation profiles for Aminbazar (4,220 tons/day) and Matuail (3,256 tons/day) through 2045.
- Calibrated the LandGEM first-order decomposition model using Bangladesh-specific parameters ($K = 0.071766$, $L_0 = 170\ m^3/ton$).
- Quantified and compared potential electricity generation through anaerobic digestion and incineration, identifying the more viable pathway based on energy-recovery potential.
- Evaluated greenhouse-gas mitigation potential and proposed engineering strategies for vertical and horizontal landfill-gas extraction systems**.**

PUBLICATIONS

Electro-thermo-mechanical coupling in functionally-graded composite laminates: A Displacement-Function approach (Manuscript is ready; Under Preparation for Submission**)**

Authors: Md Shah Wali Ullah, Tonima Islam, Mohammad Abdul Mohaimin, Md Nahid Shahriare, S Reaz Ahmed

Abstract: Reliable and efficient modeling of the coupled electro-thermo-mechanical response of functionally graded metal-matrix composite laminates is of utmost practical importance for the safe and economic design of multifunctional load-bearing structures subjected to electrical and thermal environments. In the present research, a novel displacement-function-based semi-analytical framework has been developed for investigating the multiphysics behavior of functionally graded composite laminates under current-induced electro-thermal loading. The electrical, thermal, and mechanical fields are sequentially coupled by incorporating temperature-dependent electro-thermal material properties and Joule heating effects. In the proposed formulation, the coupled thermomechanical equilibrium problem is modeled through a single scalar function, governed by a fourth-order partial differential equation of equilibrium, from which laminate strains and ply-level stresses are systematically derived. The proposed modeling scheme is applied to Al-SiC functionally graded symmetric laminates with gradually varying SiC reinforcement volume fraction across the thickness, characterized by a power-law grading function. Parametric investigations reveal that the grading profile, applied current density, and internal heat generation strongly govern the electrical potential distribution, temperature evolution, and stress development within the laminate. The predicted electro-thermo-mechanical responses are validated against finite-element simulations, showing excellent agreement and confirming the accuracy of the model. The proposed formulation provides an efficient and physically transparent framework for analyzing electro-thermo-mechanical interactions in functionally graded composite laminates and offers a viable alternative to computationally intensive conventional numerical approaches for advanced multifunctional material design.

Assessment of Landfill Gas-to-Energy Potential and Technologies at Arefin Nagar and Fatehabad Landfills Using Site-Specific LandGEM Modeling (Accepted at ICCHE BUET 2026, Presentation at December 2026**)**

Authors: Pritam Biswangri, Ayan Chowdhury, Amitav Mandal, Mohammad Abdul Mohaimin

Venue: 8th International Conference on Chemical Engineering (ICCHE 2026), BUET (2026) — Status: Accepted for Presentation

Abstract: The second-largest city in Bangladesh, Chattogram, produces over 3,000 tonnes of municipal solid waste (MSW) every day. The majority of this trash is dumped in open dumpsites at Arefin Nagar and Anandabazar without leachate treatment or gas control systems. Since organic waste makes up 62-73% of the MSW stream, anaerobic decomposition produces large amounts of landfill gas (LFG) that are now released uncontrollably into the atmosphere, posing both greenhouse gas and public health risks to nearby communities. As Chattogram City Corporation prepares to transition disposal operations to a new engineered landfill at Fatehabad by 2027, a critical opportunity exists to integrate gas-to-energy infrastructure into both the closing and incoming sites. This study projects methane generation at Arefin Nagar and Fatehabad through 2090 using the USEPA LandGEM v3.02 Site Specific-1 (SP-1) model calibrated with locally determined parameters. According to modeling data, peak methane generation was 19,647 Mg yr^{-1} at Arefin Nagar (2027) and 22,600 Mg yr^{-1} at Fatehabad (2038). From 2030 to 2041, the overall generation exceeded 20,000 Mg yr^{-1}. Converting this LFG to electricity via reciprocating IC engines yields peak recoverable outputs of 71.7 GWh/yr and 82.5 GWh/yr at the two sites, respectively, rising to over 130 GWh/yr under combined heat-and-power (CHP) operation at higher collection efficiency. Based on these findings, LFG recovery via reciprocating IC engines is identified as the most technically and economically suitable near-term energy extraction technique for Chattogram, owing to its compatibility with high-moisture organic waste, modular low-cost deployment (USD 1,000-2,500/kW), and ability to capture methane already accumulated within existing waste mass. The waste stream's high moisture content and low calorific value (3-5 MJ/kg) make incineration, pyrolysis, and gasification inappropriate. Anaerobic digestion is recommended as a complementary medium-term pathway for fresh organic waste, subject to the development of source-separation infrastructure. A phased implementation strategy is proposed, beginning with LFG well-field deployment at Arefin Nagar ahead of its 2026-2027 closure.

PROFESSIONAL EXPERIENCE

Bashundhara Oil & Gas Company Limited (BOGCL)

Assistant Engineer — Plant Inspection & Integrity **Jan 2026 – Present**
Keraniganj, Dhaka

Responsible for the mechanical integrity, inspection planning, and reliability of 250+ static equipment assets — including pressure vessels, columns, storage tanks, furnaces, boilers, heat exchangers, and process/utility/offsite piping — across a 200 TPH crude-oil refinery, ensuring full organizational compliance with API and ASME codes and standards while leading a team of four inspection/field personnel.

- **Asset Integrity Program Ownership:** Led and managed the end-to-end inspection program for 250+ static equipment assets, ensuring organizational compliance with API 510, 570, 653, 598, 576, 526, 527, 573, 571, 572, 574 and related codes, ASME BPVC Section V, Section VIII Div. 1, Section IX, and the ASME B31, B36, and B16 series.
- **Inspection Documentation Systems:** Developed and institutionalized a structured, fully traceable inspection documentation and tracking system for 250+ assets in accordance with API 510, 570, and 653, creating a centralized repository of inspection, damage, and maintenance histories that improved audit-readiness, compliance tracking, and long-term integrity governance.
- **Inspection Test Plans (ITPs):** Developed comprehensive Inspection Test Plans to standardize inspection procedures and intervals, optimizing maintenance requirements and reducing unscheduled plant downtime.
- **Data-Driven Inspection Strategy:** Drove the development of a data-driven inspection program using historical inspection data and damage-mechanism analysis to optimize inspection intervals, reduce maintenance costs, and eliminate unnecessary work.
- **Team Leadership:** Directed and mentored a team of four inspection professionals — setting performance standards, delegating technical responsibilities, and supervising daily NDE and inspection execution — fostering a culture of code compliance, quality consciousness, and operational discipline aligned with API 571 damage-mechanism awareness.
- **Risk-Based Engineering Assessment:** Oversaw risk-based assessment of inspection findings, including damage-mechanism identification (API 571), corrosion-rate analysis, and remaining-life evaluation (API 510, 570, 653), translating technical data into actionable repair, replacement, upgrade, and modification decisions.
- **NDT & Pressure Testing Programs:** Planned, coordinated, and reviewed multi-discipline NDT programs (VT, PT, MT, UT) per ASME Section V, and pressure-testing activities across diverse equipment categories per ASME Section VIII, ASME B31.3, and API 510/570, ensuring execution quality and full code conformance.
- **Pressure Safety Devices:** Owned compliance oversight for all PSV/PRV testing programs per API 526/527/576, maintaining test schedules, evaluating outcomes against code provisions, and proactively adjusting retest frequencies per API 510 and ASME Section I/VIII requirements to manage risk and regulatory exposure.
- **Field Quality Assurance:** Managed field QA activities including CUI assessments (API 583), welding inspection — pre-, in-process, and post-weld (ASME Section IX, AWS D1.1) — and coating inspection (surface preparation, profile, WFT, DFT) per SSPC/NACE standards, ensuring contractor and personnel compliance with quality and safety requirements.
- **Repair, Turnaround & Shutdown Support:** Supported major maintenance, shutdown, overhaul, and refurbishment activities by recommending suitable repair methods per applicable engineering standards and verifying repair quality through inspection, NDT, and integrity testing.
- **Root-Cause & Failure Investigation:** Drove failure investigation and risk mitigation through root-cause analysis, engineering assessments, and technical recommendations to address underlying degradation mechanisms and reduce recurrence.
- **Material Quality Verification:** Coordinated material quality inspections for supplier samples and incoming engineering materials, verifying compliance with technical specifications and liaising with Supply Chain, vendors, and user departments to safeguard capital investment.
- **Cross-Functional Coordination:** Coordinated inspection activities with Production, Mechanical Maintenance, Planning, and EHS teams to ensure equipment readiness, safe work execution, and minimal operational disruption.
- **Technical Reporting:** Prepared and delivered technical documentation — inspection reports, equipment history records, damage assessments, and engineering recommendations — to support maintenance planning and long-term asset integrity, and to enable data-driven decisions for cross-functional stakeholders.

- **Safety & Commissioning Reviews:** Represented the inspection function in pre-startup safety reviews (PSSR) per OSHA PSM/API RP 750 guidelines and major overhaul inspections, verifying equipment readiness and safe commissioning in line with applicable code, specification, and safety requirements.

Abul Khair Group

Technical Management Trainee — Service Management (Coffee Vending Machines) **May 2025 – Dec 2025**
Pahartali, Chittagong

Served as Divisional Technical Lead for coffee-vending-machine service operations across six divisions of Bangladesh, managing a fleet of 7,500+ machines and a distributed team of approximately 42 Machine Service Officers.

- **Nationwide Service Operations:** Managed technical service operations for 7,500+ coffee vending machines across six divisions of Bangladesh, improving equipment availability, service performance, and customer satisfaction.
- **Team Leadership:** Led and coordinated approximately 42 Machine Service Officers across geographically distributed regions, setting service standards and ensuring timely maintenance, operational efficiency, and maximum machine uptime.
- **Field Performance Assessment:** Conducted regular market visits to assess Service Officer performance and gather direct customer feedback, generating insights for service-process and machine-design improvements.
- **Data-Driven Reporting:** Prepared and maintained daily operational reports on pending tasks, machine breakdowns, and team performance; analyzed machine-breakdown trends and service metrics to implement data-driven improvement strategies; developed and analyzed the Customer Satisfaction Index (CSI) to evaluate and enhance service quality.
- **Spare Parts & Procurement:** Managed spare-parts inventory and coordinated international and local procurement to maintain optimum stock levels and ensure uninterrupted field service operations.
- **Cross-Functional Liaison:** Acted as liaison between Sales, Service, Logistics, and Supplier teams — as well as branding and business-development functions — to ensure smooth coordination and prompt resolution of technical and customer-related issues.

Industrial Attachment — British American Tobacco Bangladesh

Industrial Trainee, Environment, Health & Safety (EHS) Department **Jun 2024 – Jul 2024**
Dhaka Factory, Mohakhali, Dhaka

University-curriculum-required industrial attachment focused on wastewater treatment systems and factory utilities within the EHS department of BAT Bangladesh's Dhaka Factory.

- **Process Assessment:** Conducted a technical assessment of the wastewater treatment (ETP) system, identifying operational inefficiencies and proposing upgrades such as alternative-pump selection, pump-installation corrections, and automated chemical dosing to improve accuracy and performance.
- **Technical Documentation:** Developed detailed process flow diagrams (from complex P&IDs) and analytical reports on the wastewater treatment process, improving employee understanding and supporting further development and maintenance.
- **Standards Benchmarking:** Compared existing plant processes against Bangladesh Department of Environment (DoE) and international environmental standards to identify equipment gaps and recommend improvements.
- **Cross-Functional Collaboration:** Collaborated with technicians and operators to identify and resolve issues in the existing ETP setup.
- **Utilities Exposure:** Reviewed EHS maintenance, risk-assessment, and environmental policies, and gained working knowledge of factory utilities including boilers, HVAC systems, compressor plants, vacuum plants, and the power-generation house.

CERTIFICATIONS & PROFESSIONAL TRAINING

- ISO 9001:2015 Quality Management System (QMS) Internal Auditor — Certified by Bashundhara Oil & Gas Company Ltd., 2026
- ISO 14001:2015 Environmental Management System (EMS) Internal Auditor — Training by Bashundhara Oil & Gas Company Ltd., 2026
- Pressure Vessel Design, Fabrication, Inspection, and Testing as per ASME BPVC — Short Course, BUET Department of Continuing Education (DCE)

ONLINE COURSES & CONTINUED LEARNING

- CS50's Introduction to Computer Science — HarvardX (edX)
- Python for Everybody Specialization (5 courses) — HarvardX (edX)
- Introduction to Programming with MATLAB — Vanderbilt University (Coursera)
- Excel Skills for Business: Essentials — Macquarie University (Coursera)
- edX Verified Certificate for Python Basics for Data Science - edX

CORE COMPETENCIES & SKILLS

Reliability & Integrity: Mechanical Integrity Management, Condition Monitoring, Risk-Based Inspection (RBI) Planning, Damage-Mechanism Identification, Corrosion-Rate & Remaining-Life Assessment, Root-Cause Failure Analysis, Fitness-for-Service Screening

Inspection & Testing: Non-Destructive Testing/Examination (VT, PT, MT, UT), Hydrostatic & Pneumatic Pressure Testing, PSV/PRV Testing & Retest-Interval Management, CUI Assessment, Welding Inspection (pre-, in-process, post-weld), Coating Inspection (surface prep, profile, WFT/DFT), Shutdown & Turnaround Inspection, Repair-Quality Verification, Incoming-Material Quality Verification

Codes & Standards: API 510, API 570, API 653, API 598, API 526, API 527, API 560, API RP 571, RP 572, RP 573, RP 574, RP 576, RP 583, API RP 750; ASME BPVC Section I, Section V, Section VIII Div. 1, Section IX; ASME B31 Series (incl. B31.3), B36 Series, B16 Series; AWS D1.1; SSPC/NACE Coating Standards; OSHA PSM; ASTM Standards

Engineering Software: SolidWorks, AutoCAD, ANSYS Workbench, COMSOL Multiphysics, SimScale, Abaqus, HTRI

Programming & Analytics: MATLAB, Python, C, SAP (PM Module), Microsoft Excel (incl. data-driven reporting & KPI tracking)

Operations & Quality: HSE Compliance, Asset & Inventory Management, Spare-Parts Planning & Procurement Coordination, Production Planning, Technical & Inspection Reporting, Documentation & Traceability Systems, Customer Satisfaction Index (CSI) Development

Leadership & Collaboration: Team Supervision & Mentoring, Cross-Functional Coordination (Production, Maintenance, Planning, EHS, Supply Chain), Stakeholder Management, Technical Presentations

Other Tools: LaTeX, MS Office (Word, PowerPoint, Excel), Canva

LEADERSHIP & VOLUNTEER EXPERIENCE

BUET Automobile Club

Vice President **Sept 2024 – Mar 2025**

- Led and organized multiple large-scale technical events at the BUET Automobile Club Festival 2024 — one of BUET's largest student-led technical festivals — including a CAD competition with 150+ participants from 7 universities, a Poster Presentation Contest on Automobile Technology (50+ attendees), and workshops and student-reception programs engaging 200–400+ participants.
- Directed logistics, budgeting, and cross-functional coordination for the festival.

AutoMaestro — Formula Student Team, BUET

Treasurer & Core Member, Logistics / Business Plan Development **Jan 2022 – Mar 2025**

- Co-developed the commercialization strategy and Business Plan Presentation for the team's Formula car, achieving 49 out of 75 points in the Business Plan Event at Formula Bharat 2023 (Combustion Class).
- Managed budgeting and sponsorship acquisition, presenting partnership proposals to industry organizations.
- Managed team social-media presence on Instagram and Twitter, increasing engagement.

Class Representative — ME Batch 19, Section A, BUET

Student-Faculty Liaison **Feb–Sep 2021, Jun–Oct 2022**

- Volunteered as class representative for two semesters, facilitating communication between faculty and students.

International Astronomical Search Collaboration (IASC) Campaign

Citizen-Science Researcher (NASA-affiliated)

- Participated in an asteroid search campaign organized by the International Astronomical Search Collaboration and NASA. During this campaign, our team discovered four new asteroids.
- This experience, my first in a collaborative setting, was invaluable. I developed teamwork skills, learned to maintain positive relationships within a team, and gained extensive knowledge about asteroids and deep space.

Badhan — BUET Zone (Sher-e-Bangla Hall Unit)

Financial Secretary

- Supported emergency blood-donor mobilization for a student-led humanitarian network, coordinating voluntary donors for 10+ patients and managing funds for operations and programs.
- Organized awareness campaigns and orientation programs for newly admitted students to encourage long-term voluntary blood-donor participation.